As described in §11.4 neutrons interact with the nucleons in the nucleus. On the other hand photons mainly interact with the electrons surrounding the nucleus (which number is proportional to the atom number) The probability of scattering of neutrons without energy change (*coherent scattering*) is approximately proportional to the area of the nucleus. As a consequence, coherent scattering of neutrons is less dependent on the atomic number than the scattering of X-rays which is proportional to the electron density (i.e. $\propto Z^2$). Thus it is clear that for a compound consisting of both heavy and light atoms, the position of the lighter atoms can be more easily determined using neutron diffraction, while X-ray diffraction is better for locating the heavier atoms. Neutron diffraction is, therefore, valuable for complementing the information obtained on the position of heavy atoms by X-ray diffraction. Neutron diffraction is particularly valuable in the location of hydrogen atoms in organic and biological materials.

11.6. MODELS FOR NUCLEAR REACTIONS

No single model is successful in explaining all the aspects of the various types of nuclear reactions. Let us consider three models which have been proposed for explaining the results of nuclear reaction studies.

11.6.1. The optical model

In the process of elastic scattering the direction of the particles is changed but none of the kinetic energy is converted to nuclear excitation energy. This would indicate that the reaction is independent of the internal structure of the nucleus and behaves much like the scattering of light from a crystal ball. Consequently, a model has been developed based on the mathematical techniques used in optics. Light shining on a transparent crystal ball is transmitted with some scattering and reflection but no absorption. Light shining on a black crystal ball is all absorbed and there is no transmission or scattering. In nuclear reactions the incoming particles are scattered in elastic scattering and are absorbed in induced transmutations. Therefore, if the nucleus is to act as a crystal ball it can be neither totally transparent nor totally black. The optical model of the nucleus is also known as the *cloudy crystal ball model*, indicating that nuclei both scatter and absorb the incoming particles.

The nucleus is described as a potential well containing neutrons and protons. The equation for the nuclear potential includes terms for absorption and scattering. This potential can be used to calculate the probability for scattering of incident particles and the angular distribution of the scattering. The model is in excellent agreement with experiments for scattering. Unfortunately, this model does not allow us to obtain much information about the consequences of the absorption of the particles which lead to inelastic scattering and transmutation.

11.6.2. Liquid-drop model

As the excitation energy of an excited nucleus increases, the energy levels get closer together. Eventually, a continuum is reached where the density of nuclear levels is so great that it is no longer possible to identify individual levels. (This is similar to the case for electronic energy levels of atoms.) When the excited nucleus emits a proton or neutron while in the continuum energy, the resultant nucleus may be still sufficiently energetic that it remains in the continuum region.

N. Bohr offered a mechanism to explain nuclear reactions in nuclei which are excited into the continuum region. When a bombarding particle is absorbed by a nucleus, the kinetic energy of the bombarding particle plus the binding energy released by its capture provides the excitation energy of the compound nucleus. In this model, the *compound nucleus* becomes uniformly excited in a manner somewhat analogous to the warming of a small glass of water upon addition of a spoonful of boiling water. As the nucleons move about and collide in the nucleus, their individual kinetic energies vary with each collision just as those of molecules in a liquid change in molecular collisions. As this process continues, there is an increase in the probability that at least one nucleon will gain kinetic energy in excess of its binding energy (assuming the total excitation energy to be larger than the binding energy). That nucleon is then evaporated (i.e. leaves the nucleus) analogously to the evaporation of molecules from liquid surfaces.

The evaporation of the nucleon decreases the excitation energy of the residual nucleus by an amount corresponding to the binding energy plus the kinetic energy of the released nucleon. The evaporation process continues until the residual excitation energy is less than the binding energy of a nucleon. The excitation energy remaining at this point is removed from the nucleus by emission of γ-rays.

Assume that the compound nucleus $^{188}_{76}$Os is formed with a total excitation energy of 20 MeV. If the average binding energy of a neutron is 6 MeV and if each neutron leaves with 3 MeV of kinetic energy, evaporation of a neutron de-excites the nucleus by 9 MeV. Therefore evaporation of two neutrons would leave the residual ^{186}Os nucleus with an excitation energy of only 2 MeV. Since this is below the binding energy of a neutron, further evaporation is not possible and γ-ray emission removes the final 2 MeV. If the ^{188}Os compound nucleus was formed by α-bombardment of ^{184}W, the reaction is represented as

$$^{184}_{74}\mathrm{W} + \alpha \rightarrow \left[^{188}_{76}\mathrm{Os}^{*}\right] \rightarrow {}^{186}_{76}\mathrm{Os} + 2\mathrm{n} \tag{11.13}$$

The "evaporation" of neutrons is also used in the terminology for the production of superheavy elemens using "cold fusion" (few neutrons "evaporate") and "hot fusion" (many neutrons "evaporate"), cf Ch 11.

11.6.3. Lifetime of the compound nucleus

An important assumption of the compound nucleus theory is that the nucleon system is held together long enough for the energy to be shared by all nucleons. Furthermore, it is assumed that the time it takes for the accumulation on one nucleon of enough energy to allow evaporation is even longer by nuclear standards. This time is of the order of 10^{-14} s as compared to a time of 10^{-20} s required for a nucleon to cross the nuclear diameter once. Since the time is so long and there are so many inter-nucleon collisions, the nucleus retains no pattern ("no memory") of its mode of formation, and the mode of decay should therefore be independent of the mode of formation and only depend on the amount of excitation energy, the *nuclear temperature* (usually expressed in MeV). For example, ^{150}Dy ($t_{½} = 7.2$ min) can be formed in the following two ways:

$$\left.\begin{matrix} {}^{12}_{6}\mathrm{C} + {}^{144}_{60}\mathrm{Nd} \\ {}^{20}_{10}\mathrm{Ne} + {}^{136}_{56}\mathrm{Ba} \end{matrix}\right\rangle \rightarrow [{}^{156}_{66}\mathrm{Dy}^{*}] \rightarrow {}^{150}_{66}\mathrm{Dy} + 6\mathrm{n} \qquad (11.14)$$

both form the excited ^{156}Dy*. To a first approximation, the probability for formation of ^{150}Dy is dependent only on the excitation energy of the compound nucleus ^{156}Dy* but

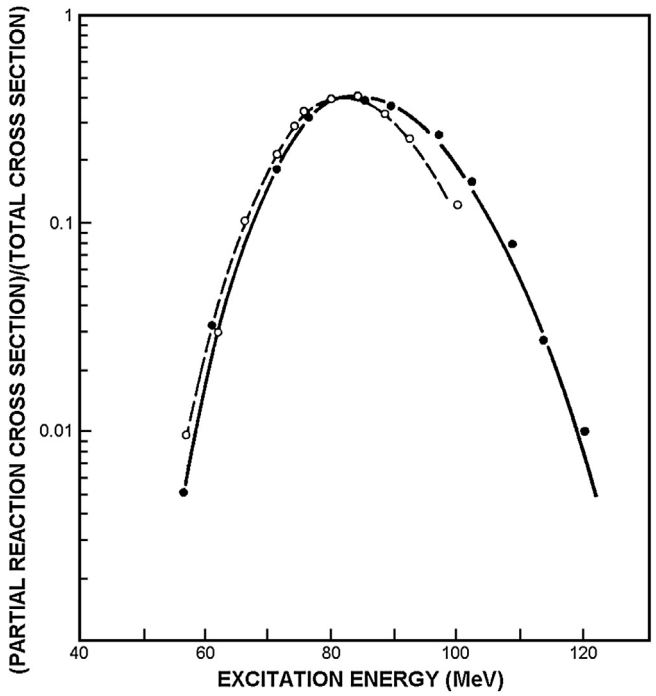


Figure 11.5 Excitation functions for the formation of ^{150}Dy, through bombardment of either ^{144}Nd with ^{12}C (○) **or** ^{136} **Ba with** ^{20}**Ne (•). (From Lefort)**

not on the manner in which this compound nucleus is formed (Fig. 11.5). This assumption of no memory of the mode of formation is not valid if very different amounts of angular momenta are involved in different modes of formation or the collision energy is very high. For example, for proton induced reactions, the excitation energy is essentially all available for internal (nucleon) excitation. By contrast, as heavier bombarding particles are used, the average angular momentum of the compound system increases and the excitation energy is divided between the rotation and internal (nucleon) excitation. The modes of subsequent decay of the compound nucleus is affected by the amount of excitation energy that was involved in the angular momentum of the compound nucleus.

In general, neutron emission is favoured over proton emission for two reasons. First, since there are usually more neutrons than protons in the nucleus, a neutron is likely to accumulate the necessary evaporation energy before a proton does. Second, a neutron can depart from the nucleus with a lower kinetic energy – the average neutron kinetic energy is 2–3 MeV. On the other hand, evaporating protons must penetrate the Coulomb barrier, so they often need about 5 MeV above their binding energy. It takes, as an average, a longer time for this amount ($\sim$12 MeV *vs.* $\sim$9 MeV for neutrons) to be concentrated on one nucleon.

Obviously, such a simple picture ignores a large number of complicating effects that can, in particular cases, reverse the order of these cross-sections. Nevertheless, despite its simplicity, the compound nucleus theory has been of great value in explaining many aspects of medium energy nuclear reactions (i.e. $\leq$ 10 MeV per nucleon of the bombarding particle).

11.6.4. Direct interaction model

The compound nucleus theory assumes that the bombarding projectile interacts with the nucleus as a whole. The nucleus is excited uniformly and evaporation of low energy nucleons follows. This model fails to explain some of the phenomena observed as the kinetic energy of the bombarding particle increases. One such observation is the occurrence of high energy neutrons and protons among the emitted particles. Another is the large cross-sections for reactions such as $X_1(p, pxn)X_2$ at energies where 6 or 7 nucleons are evaporated in order to de-excite the nucleus. At still higher energies compound nucleus formation is too slow to occur at all and the target nucleus (and also projectile in case of heavy ions) splits rapidly into several fragments.

Figure 11.6 shows the cross-section for production of nuclides of $A = 20 - 200$ when ^{208}Pb is bombarded with protons of energies 40, 480, and 3000 MeV. Such reactions, yielding a large number of products at high projectile energies, have been extensively studied by nuclear chemists, partly because the mixture of products required separation by radiochemical techniques. Although some ambiguity exists in its use, the term *spallation* is often used for reactions in which a number of particles are emitted as a result of a direct interaction. At very high energies, not only is a broad range of products

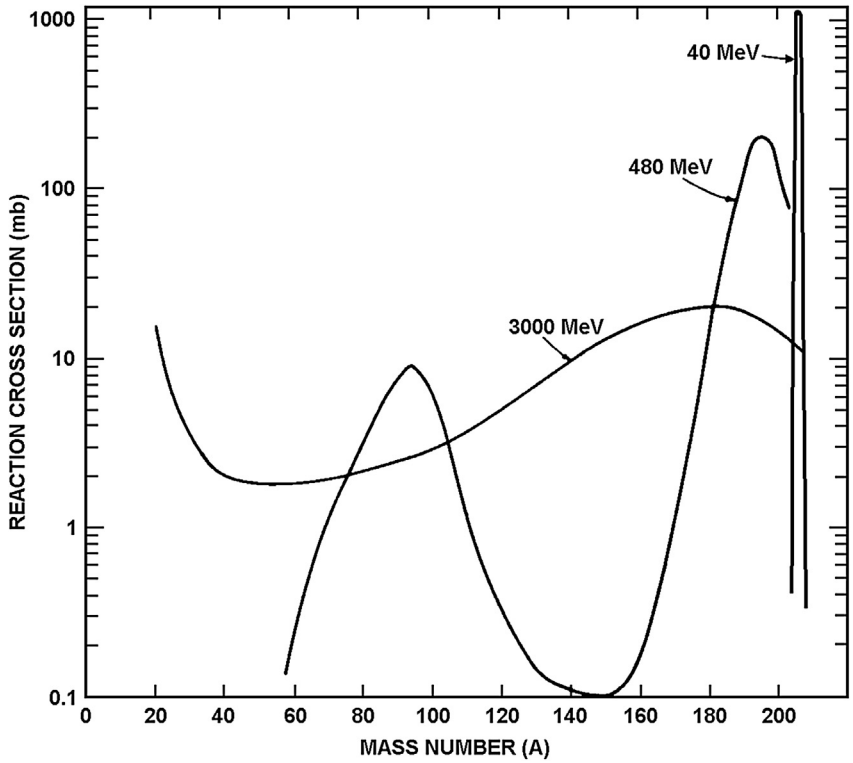


Figure 11.6 Mass yield curve obtained by bombardment of lead with high energy protons. (From Miller and Hudis.)

formed but the probability for the formation of these products is, within an order of magnitude, similar for every mass number, except around the projectile and target masses where strong peaks occur (Fig. 11.7). Further, studies at bombarding energies above 100 MeV/u show that high energy protons, neutrons, and heavier particles are emitted from the nucleus in a forward direction. Compound nucleus evaporation would be expected to be isotropic (i.e. show no directional preference in the center-of-mass system).

Serber has suggested a mechanism that satisfactorily accounts for many features of nuclear reactions at bombardment energies above 50 MeV for protons, deuterons, and α-particles. At such energies the relative speed between projectile and target nuclei is so high (near **c**) that the time available for distribution of energy between all nucleons is too short and we can initially consider projectile and target nuclei as consisting of fairly isolated nucleons. The concept of a common nuclear temperature is no longer valid. He proposed that high energy reactions occur in two stages.

(i) During the first stage the nucleons in the incoming particle undergoes direct collision with individual target nucleons. In these collisions the struck nucleon

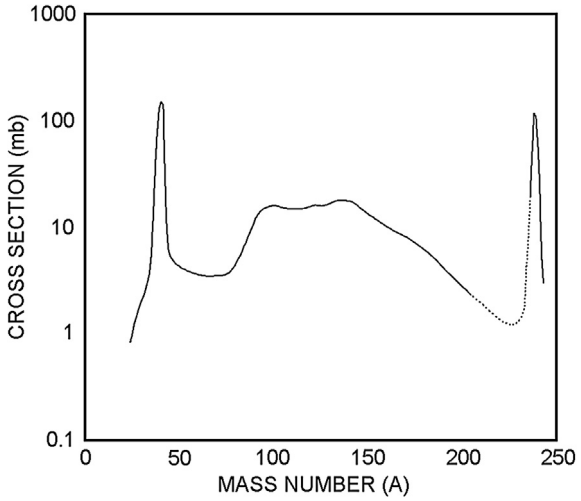


Figure 11.7 Mass yield curve obtained by bombardment of ^{238}U with ^{40}Ar-ions at 7.2 MeV/u. (From Kratz, Liljenzin, Norris, and Seaborg.)

often receives energy much in excess of its binding energy. Consequently, after each collision both the nucleon belonging initially to the bombarding particle and the struck nucleon have some probability of escaping the nucleus since their kinetic energies are greater than their binding energies. If both particles escape, the nucleus is usually left with only a small amount of excitation energy. This explains the high cross-section for (p,pn) reactions. In support of this explanation, both the emitted proton and neutron have large kinetic energies. Either one or both of the original pair may collide with other nucleons in the nucleus rather than escape. During this initial stage, known as the *knock-on-cascade process*, the total number of direct collisions may be one or many. After a period lasting about 10^{-19} s, some of the struck nucleons have left the nucleus.

(ii) In the remaining nucleus the residual excitation energy is uniformly distributed. The reaction then enters its second and slower stage, during which the residual excitation energy is lost by nucleon evaporation. This stage resembles the compound nucleus process very closely.

In Figure 11.6 at 40 MeV only the second, evaporation, stage is observed as seen by the narrow mass distribution curve. The curves for 480 and 3000 MeV reflect the increased importance of the first, direct interaction stage which leads to a broad spectrum of product mass numbers.

The experimental data have been successfully reproduced using a calculation technique known as the Monte Carlo method and assuming a Fermi gas model for the nucleus. This model treats the nucleons like molecules of a very cold ideal gas in a potential well. The nucleons do not follow the Pauli exclusion principle and fill all vacant orbitals.

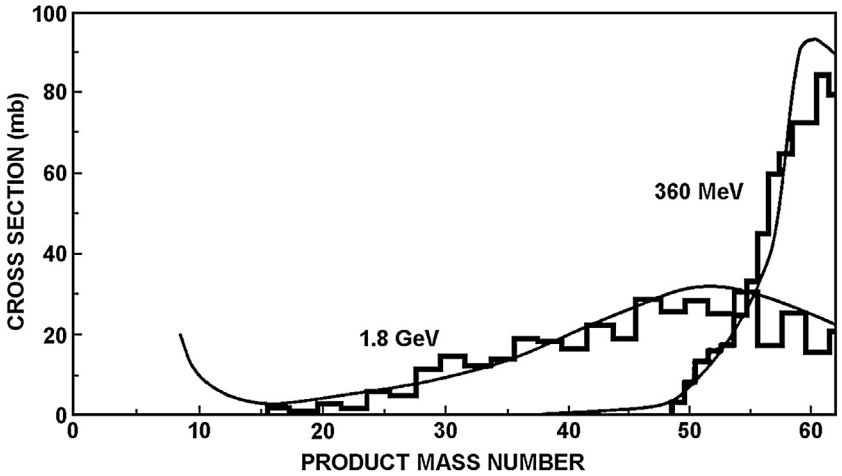


Figure 11.8 Comparison of mass yield curves for proton interaction with copper as predicted by Monte Carlo calculations (histogram) with experimental results (curve). (From G. Friedlander.)

In a *Monte Carlo calculation*, the history of each incident nucleon is studied in all its collisions in the nucleus. Each collision with a target nucleon is characterized by probability distributions for occurrence, for energy, for angular distribution, and for pion formation. The outcome of each process is obtained from a series of properly distributed random numbers and scores are accumulated for each process. The calculations are repeated for different impact points. It is the use of random numbers that is the basis of the Monte Carlo technique. Modern computers allow so many random number calculations that a useful pattern of events emerges from the cumulative scores.

Figure 11.8 shows the quite satisfactory agreement for two bombardment energies between experimental yields of A in proton bombardment of copper (solid lines) and a Monte Carlo calculation (histograms).

11.7. NUCLEAR FISSION

Nuclear fission can be either spontaneous or induced by particle (e.g. neutron) bombardment. It is fairly unique among nuclear reactions since the nucleus divides into a few heavy parts with the release of a large amount of energy, about 200 MeV per fission. It is probably no overstatement to say that fission is the most important nuclear process among man made ones, both for its potential to destroy civilization through the use of weapons and its potential through reactors to supply abundant power for all people.

An aspect of fission which has been studied extensively is the distribution of mass, charge and kinetic energy among the fragments formed in fission. No matter how nuclei are made to undergo fission, fragments of various masses are formed which result in production of chemical elements as light as zinc (atomic number 30) and as heavy as